

Generation Interconnection Request System Impact Study Report

For Queue Project Q19-1333

Aguada 7201 138 kV
November 30, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1333. The Interconnection Customer proposes a 399.9 MW solar photovoltaic facility interconnecting at the Aguada 7201 138 kV substation in the West Virginia area. The request was received on November 10, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 493 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 24 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$131,423,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Waterton / Iron Grove 138 kV	Rockaway Beach / Mattoon E 138 kV	204.8	177	11.4
Goat Rock / Panda 138 kV	Panda / Northern Steel Casting 138 kV Line	204.8	188	8.7
Waterton / Hampshire Paper 138 kV	Mattoon E / Rockaway Beach 138 kV	200.6	177	11.4
Northern Steel Casting / Goat Rock 138 kV	Panda / Northern Steel Casting 138 kV	196.3	199	8.7
Northern Steel Casting / Panda 138 kV line	Goat Rock / Northern Steel Casting 138 kV	191.1	226	9.7
Carlsbad Waterfield / Menlo 138 kV	Menlo / W Woodland 138 kV	185.8	205	8.7
Texas Farms Tap / Lemon Gulch 138 kV	Aguada 7201 Substation / Union Valley 138 kV	176.7	221	100.0
Aguada 7201 / Lemon Gulch 138 kV	Aguada 7201 / Union Valley 138 kV	174.7	225	100.0
Aguada 7201 / Union Valley 138 kV L	Lemon Gulch / Aguada 7201 138 kV Line	171.2	230	100.0
Carlsbad Waterfield / E. Ripley 138 kV	Menlo / W Woodland 138 kV L	171.0	219	8.7
Union Valley Substation / Iron Grove 138 kV L	Aguada 7201 / Lemon Gulch 138 kV	169.6	204	54.4
E. Ripley / W Fork 138 kV L	Menlo / West Woodlnd 138 kV Line	161.1	227	8.7
Menlo / Panda 138 kV	Dillard-Orange / Fishkill Plains 345 kV Line	127.1	383	10.5
Solana / Northern Steel Casting 138 kV Line	Fishkill Plains / Dillard-Orange 345 kV L	125.7	397	10.5

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Solana / Hampshire Paper 138 kV	Fishkill Plains Substation / Dillard-Orange 345 kV L	125.2	403	10.5
Eules / Tank Hill Substation 138 kV	Menlo / Carlsbad Waterfield 138 kV	124.6	230	7.0
Eules / West Woodland 138 kV Line	Carlsbad Waterfield / Menlo 138 kV line	124.4	233	7.0
Menlo Substation / W Woodland 138 kV	Carlsbad Waterfield / Menlo 138 kV	123.8	237	7.0
N Adams / Radisson 138 kV line	Aguada 7201 / Union Valley 138 kV line	119.4	193	56.7
Amboy / Eastshore 138 kV	Texas Farms Tap Substation / Radisson 138 kV	118.4	152	36.2
Radisson / Texas Farms Tap 138 kV	Aguada 7201 / Union Valley 138 kV	116.0	202	56.7
E Sidney Substation / Whitefish 138 kV	Fishkill Plains / Dillard-Orange 345 kV	105.8	267	10.2
E Sidney / Semiahmoo 138 kV	Fishkill Plains / Dillard-Orange 345 kV	105.4	267	10.2
Eastshore Substation / Dillard-Orange 138 kV	Radisson Substation / Texas Farms Tap 138 kV	104.6	170	36.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Waterton / Iron Grove 138 kV	Rebuild line with higher-capacity conductor	\$3,652,000	92% (\$3,344,000)
Panda / Goat Rock 138 kV Line	Rebuild line with higher-capacity conductor	\$11,796,000	89% (\$10,488,000)
Waterton / Hampshire Paper 138 kV	Rebuild line with higher-capacity conductor	\$4,618,000	92% (\$4,229,000)
Goat Rock Substation / Northern Steel Casting 138 kV line	Rebuild line with higher-capacity conductor	\$6,383,000	89% (\$5,675,000)
Northern Steel Casting / Panda 138 kV line	Rebuild line with higher-capacity conductor	\$2,925,000	89% (\$2,600,000)
Carlsbad Waterfield / Menlo 138 kV	Rebuild line with higher-capacity conductor	\$3,443,000	91% (\$3,138,000)
Texas Farms Tap / Lemon Gulch Substation 138 kV	Rebuild line with higher-capacity conductor	\$5,961,000	100% (\$5,961,000)
Lemon Gulch / Aguada 7201 138 kV	Rebuild line with higher-capacity conductor	\$3,515,000	100% (\$3,515,000)
Aguada 7201 / Union Vly 138 kV	Rebuild line with higher-capacity conductor	\$16,781,000	100% (\$16,781,000)
E. Ripley / Carlsbad Waterfield 138 kV	Rebuild line with higher-capacity conductor	\$5,500,000	91% (\$5,013,000)
Union Valley Substation / Iron Grove 138 kV L	Rebuild line with higher-capacity conductor	\$5,828,000	100% (\$5,828,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
E. Ripley / W Fork 138 kV L	Rebuild line with higher-capacity conductor	\$4,243,000	91% (\$3,867,000)
Menlo / Panda 138 kV	Rebuild line with higher-capacity conductor	\$7,123,000	90% (\$6,381,000)
Solana / Northern Steel Casting 138 kV Line	Rebuild line with higher-capacity conductor	\$9,149,000	90% (\$8,196,000)
Solana / Hampshire Paper 138 kV L	Rebuild line with higher-capacity conductor	\$7,163,000	90% (\$6,417,000)
Euless / Tank Hill Substation 138 kV	Rebuild line with higher-capacity conductor	\$6,022,000	85% (\$5,142,000)
Euless / West Woodland 138 kV line	Rebuild line with higher-capacity conductor	\$3,704,000	85% (\$3,162,000)
Menlo Substation / W Woodland 138 kV	Rebuild line with higher-capacity conductor	\$5,562,000	85% (\$4,749,000)
N Adams / Radisson 138 kV line	Rebuild line with higher-capacity conductor	\$5,777,000	100% (\$5,777,000)
Eastshore / Amboy 138 kV	Rebuild line with higher-capacity conductor	\$4,759,000	100% (\$4,759,000)
Raisson / Texas Farms Tap 138 kV	Rebuild line with higher-capacity conductor	\$7,847,000	100% (\$7,847,000)
E Sidney Substation / Whitefish 138 kV	Reconductor line	\$3,094,000	71% (\$2,183,000)
E Sidney / Semiahmoo 138 kV	Reconductor line	\$3,407,000	71% (\$2,404,000)
Eastshore Substation / Dillard-Orange 138 kV	Reconductor line	\$3,967,000	100% (\$3,967,000)
	Total allocated to project		\$131,423,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.